Class 3

Spatial Data Collection

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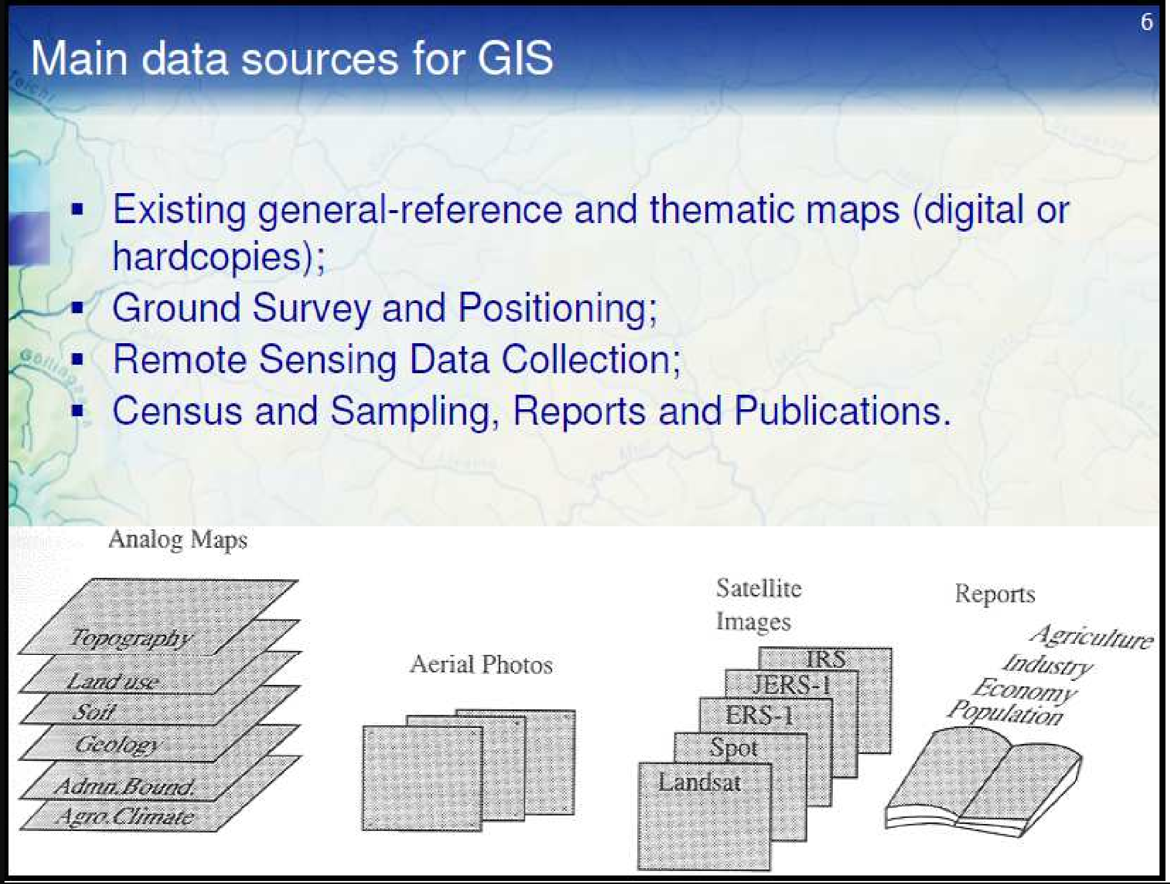
August 14th, 2026

GIS Layer contents

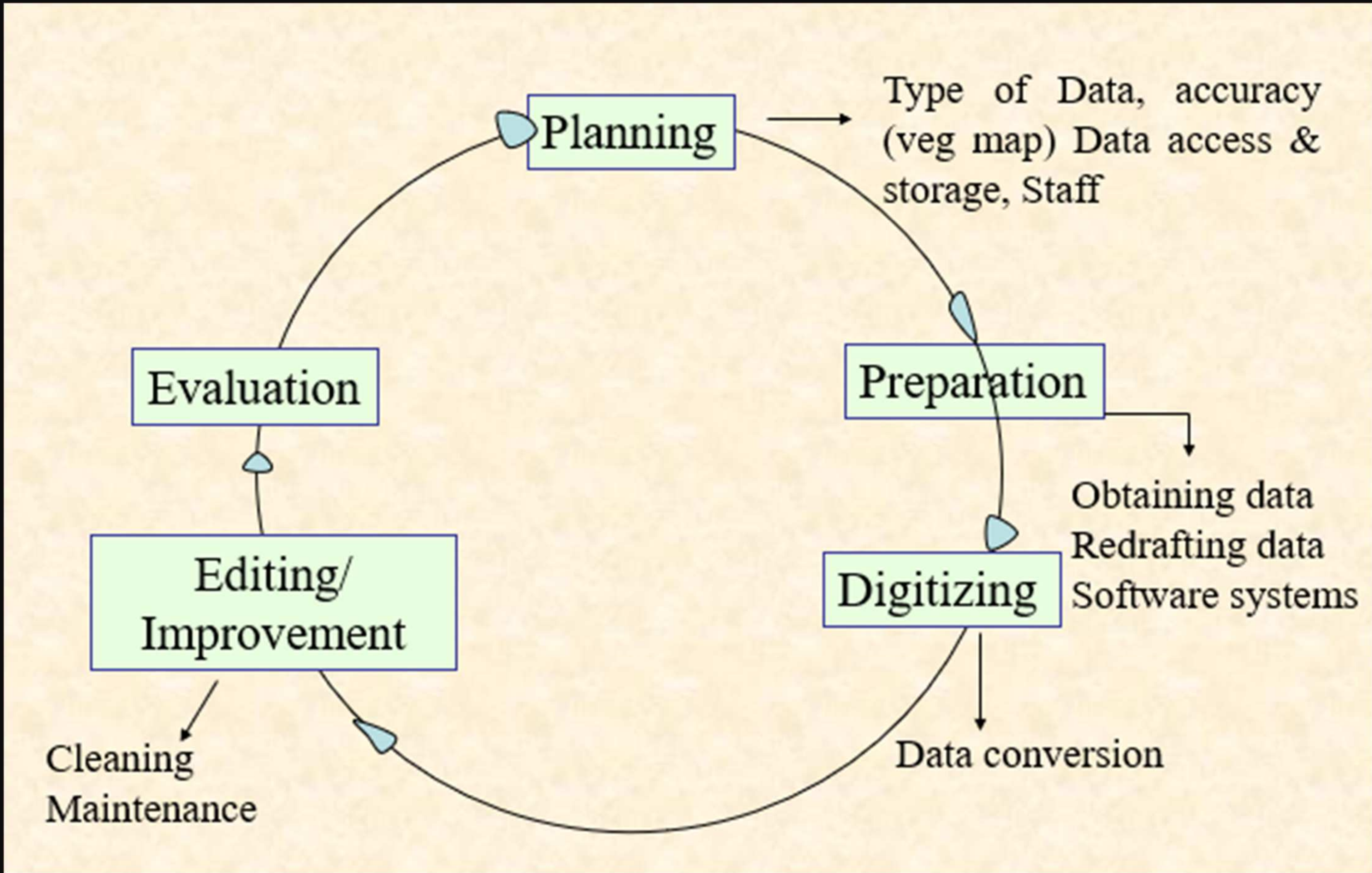
- *Spatial*, *Coordinate System* info, *Attribute*, *Metadata* , *Symbology*



Data collection

- Either **digital** or **analog**.
- "***While models which uses the data are important to support decision making activities, a large portion of the investment will be in obtaining, converting and storing new data" – Kennedy and Guinn (1975***)
- Data collection still remains a time consuming, tedious, and expensive process. Usually it accounts for **15-50%** of the total cost of a GIS project.

Data collection workflow



GIS Data Collection - Types

One of the most time-consuming and expensive, yet important of GIS tasks.

many *diverse sources of geographic data* and

many *methods available to enter them into a GIS*.

Two broad types of data collection:

- ***Data capture*** (direct collection)
- **Two broad capture methods**
  - **primary**: direct measurement;
  - **secondary**: indirect derivation from other sources (those reused from earlier studies)
- ***Data transfer***

  (importing digital data from other sources)

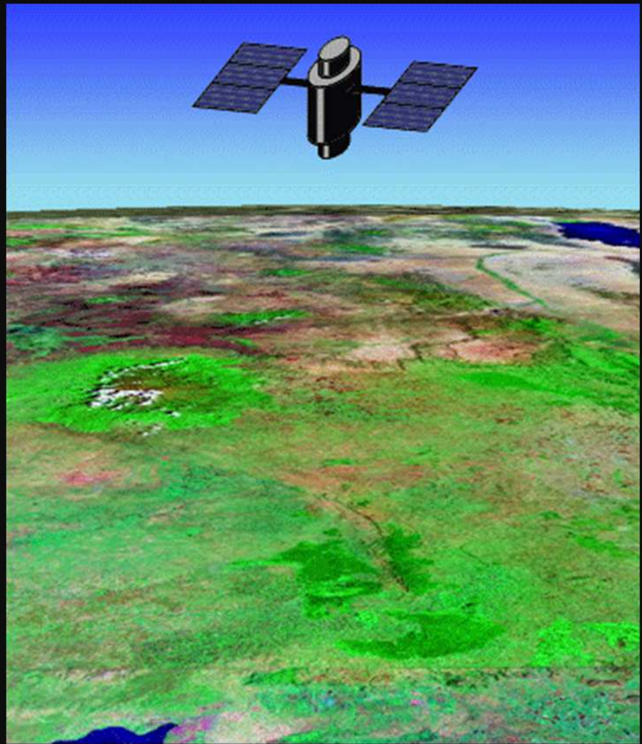
Capturing attribute data

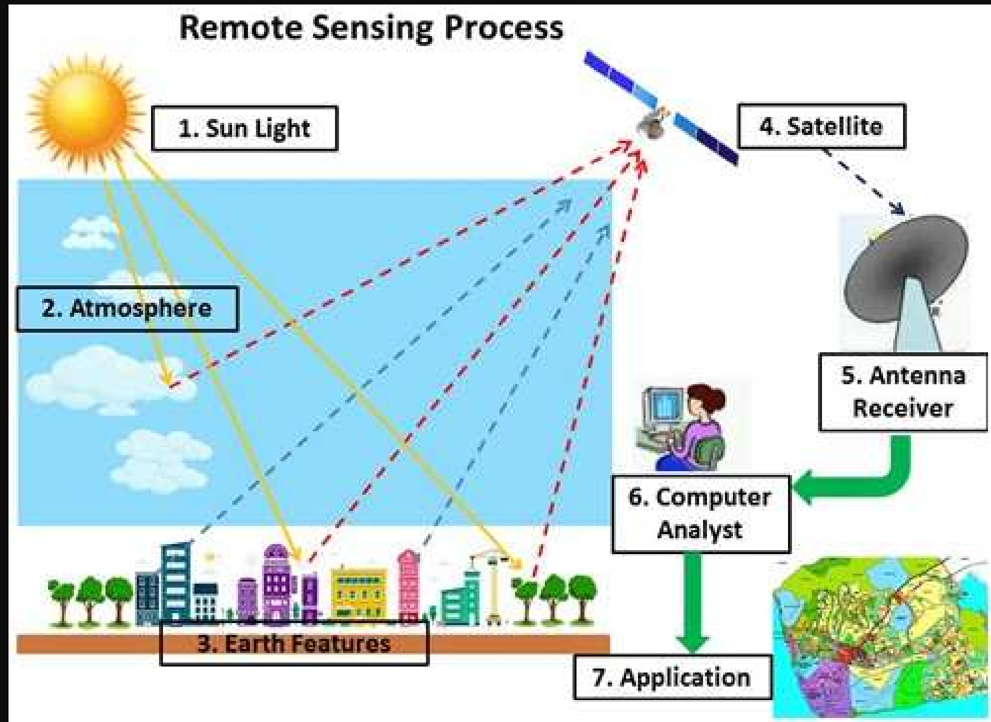
Vector and raster data capture techniques

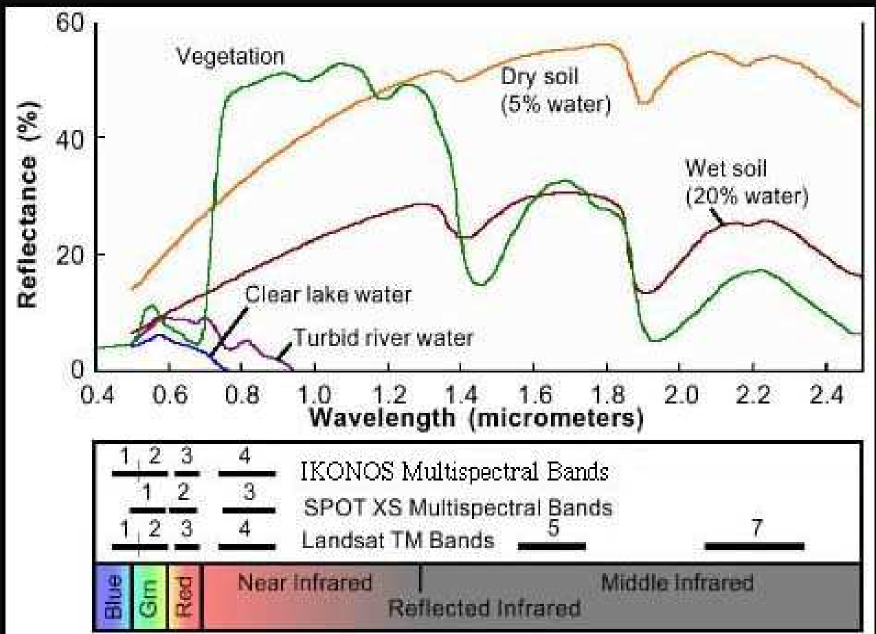
	Raster	Vector
Primary	**Digital remote sensing images**	**Survey measurements**
	Digital aerial photographs	**GPS measurements**
Secondary	**Scanned maps** **Photographs**	**Topographic surveys**
	Any other method ?	

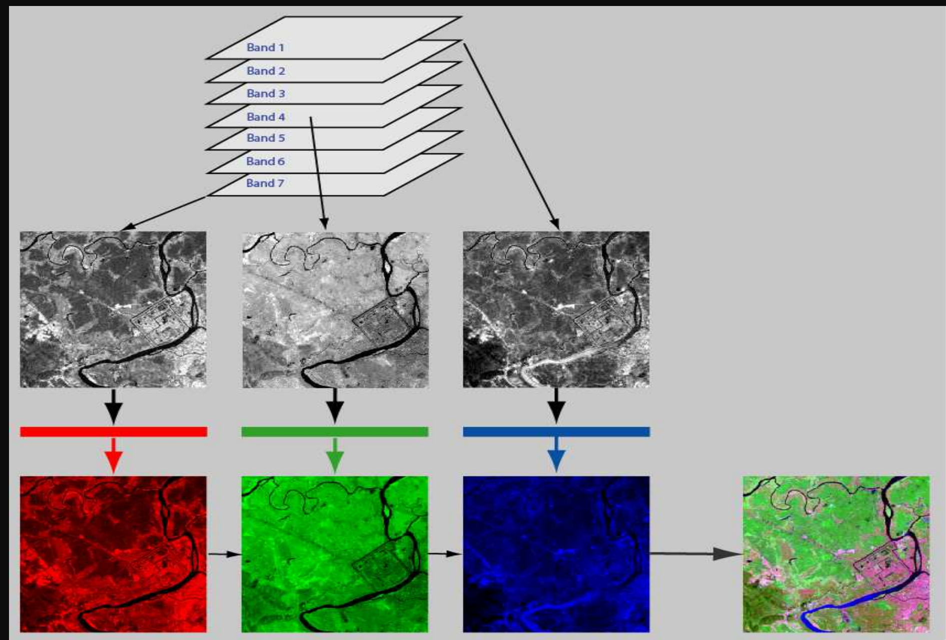
Attribute data →

Remote sensing









BLUE
(0.4-0.5 μm)

GREEN BAND
(0.5-0.6 μm)

TRUE COLOUR COMPOSITE

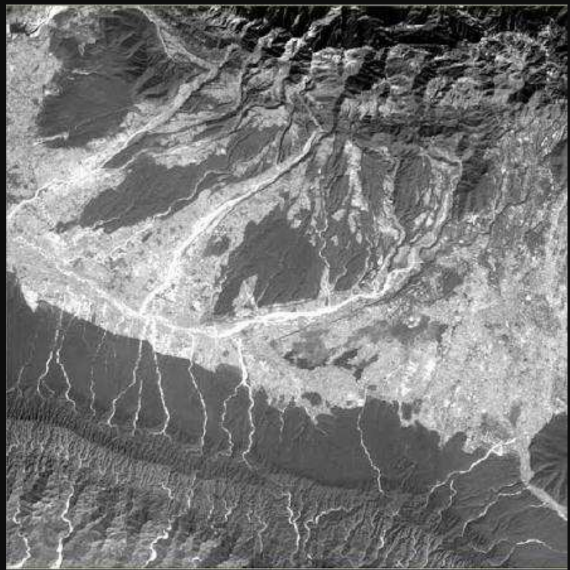
RED BAND
(0.6-0.7 μm)

NEAR IR
(0.7-0.9 μm)

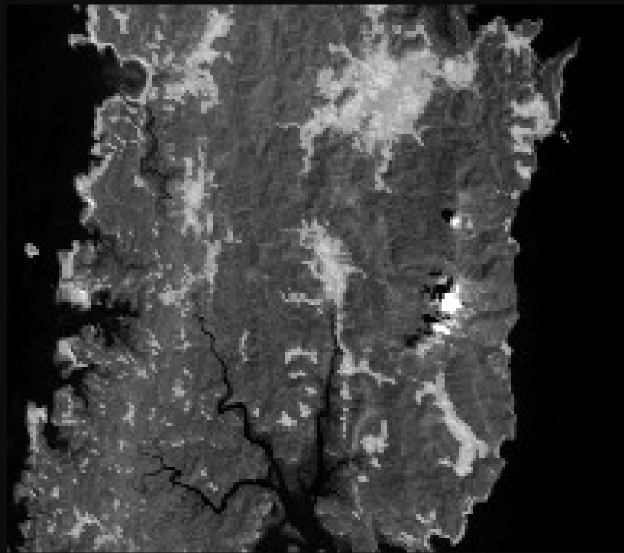
FALSE COLOUR COMPOSITE

1 : SAND 2 : VEGETATION 3 : WATER

Spectral and Spatial Resolution



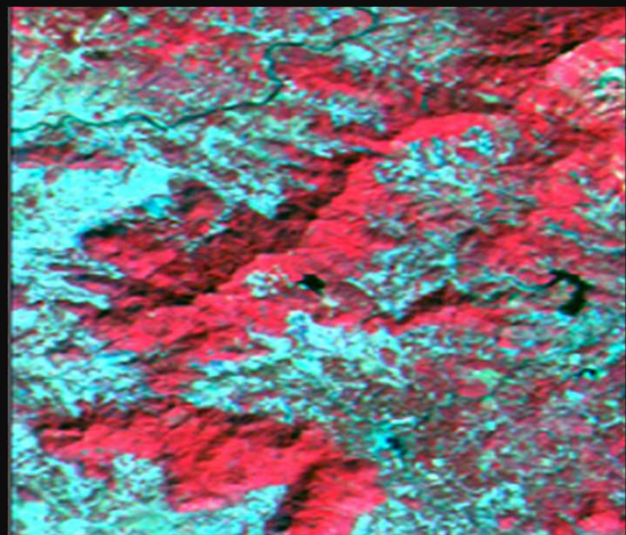
500-850nm
Cartosat-I



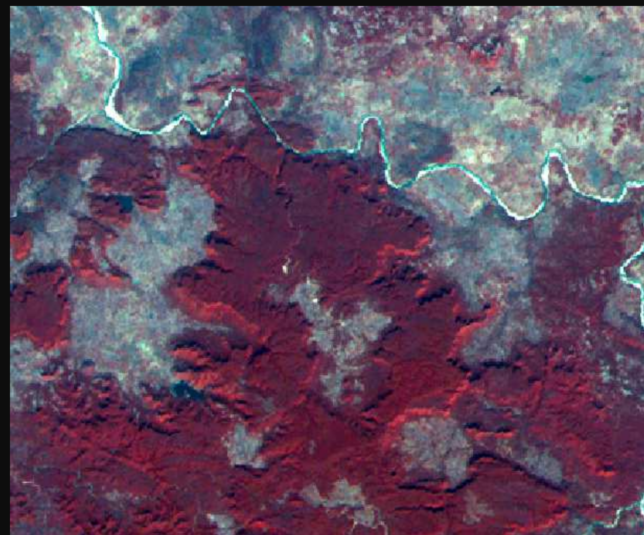
500-750nm
PAN

GRIR
LISS-IV

GRN-MIR
LISS III



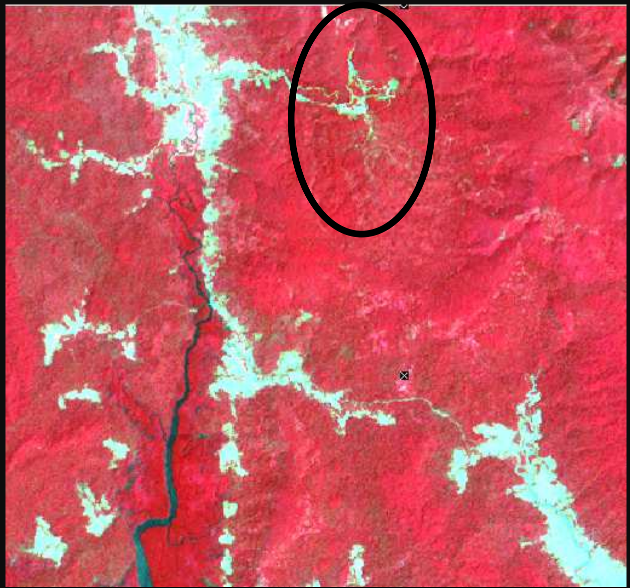
GRN-MIR
AWiFS



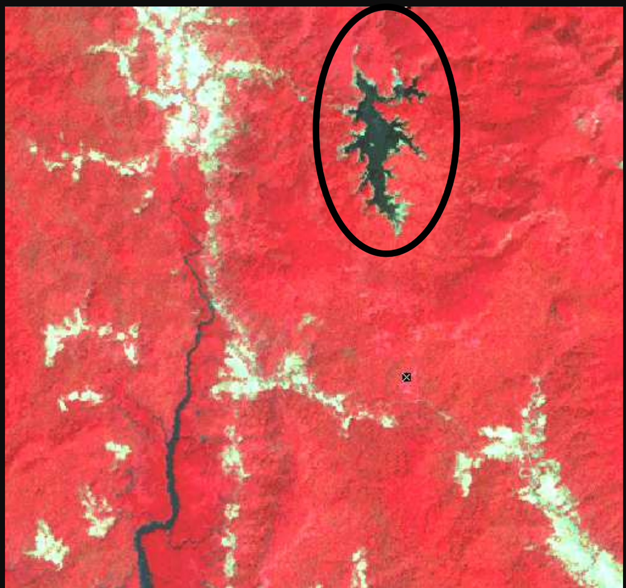
BGRIR
LISS I

Temporal Resolution

Andaman Islands

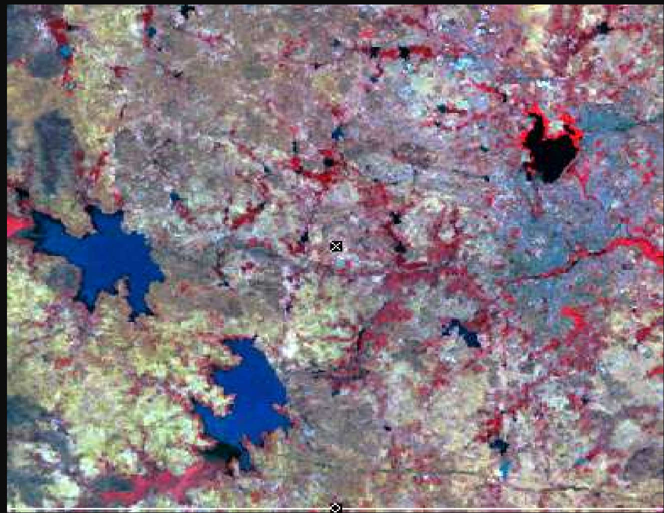


LISS-III March 1999

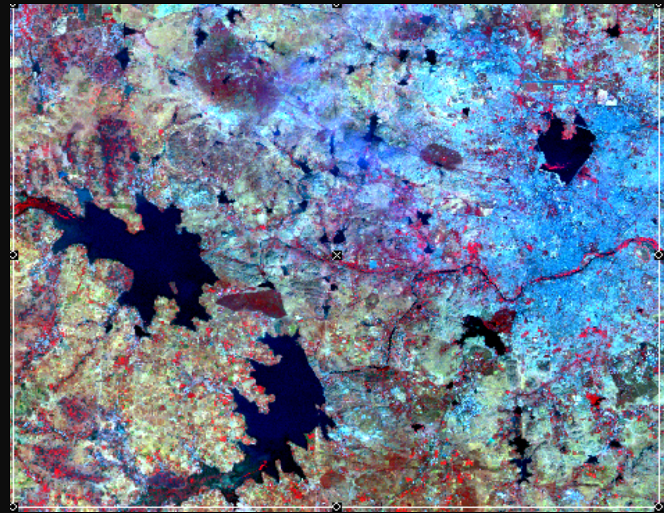


LISS-III Feb 2005

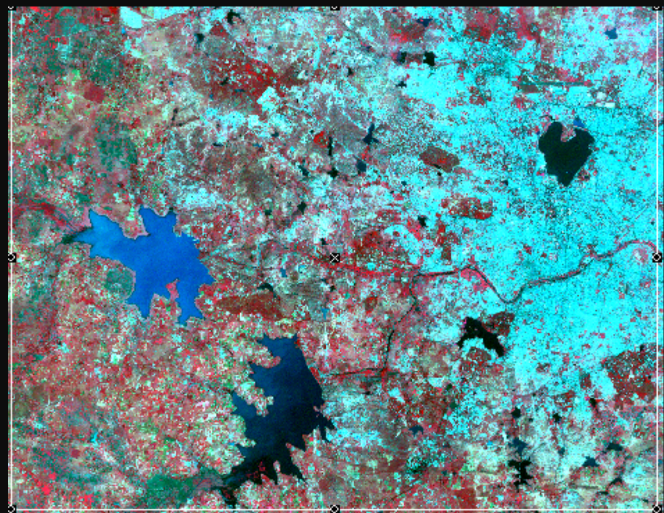
Hyderabad city



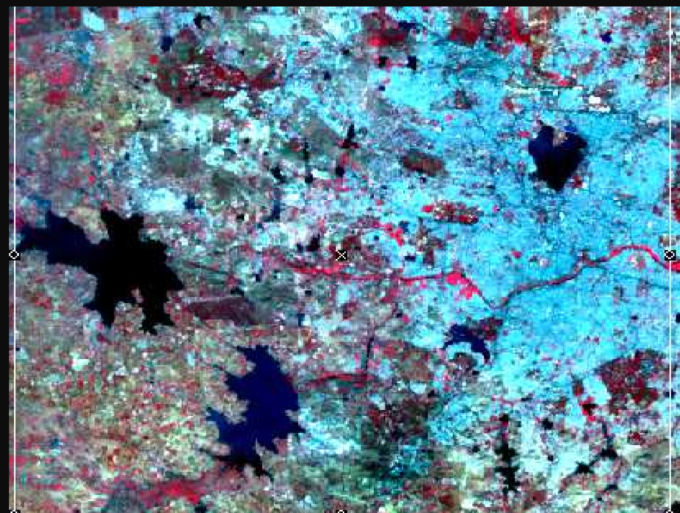
MSS -1976



TM -1989



ETM -2001



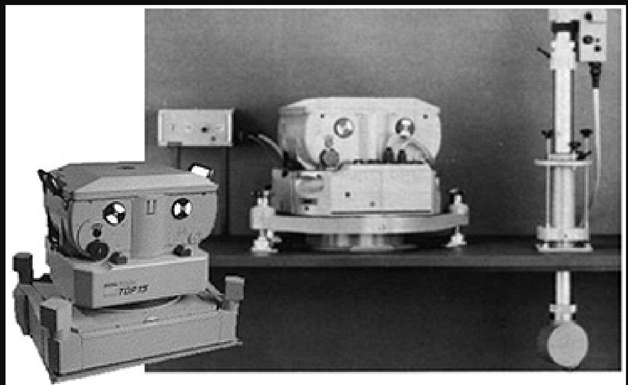
AWiFS -2001

Wide utility of RS data is mainly due to…..

Aerial Photography

SR and AP are technically similar; Difference in Capturing and Interpretation

AP normally collected using analog optical camera, later rasterized by scanning film negative.



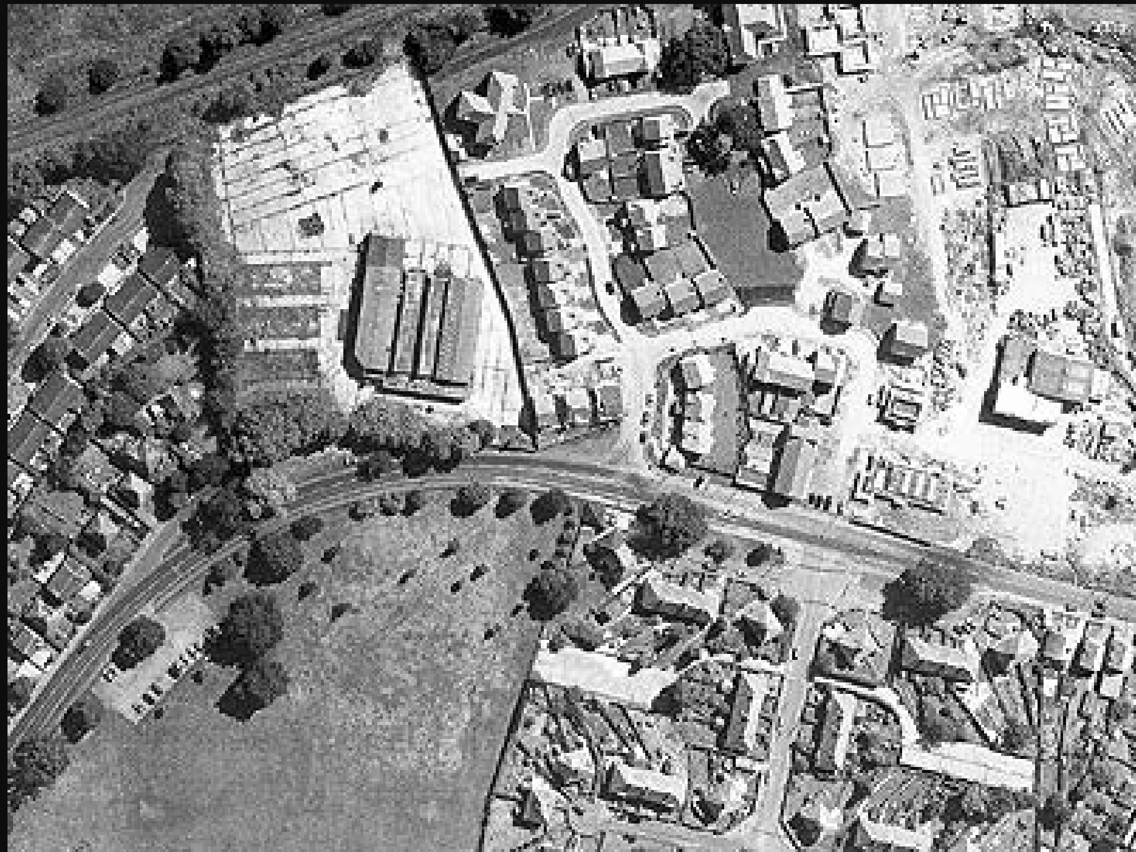
Cameras mounted in the **nose or underbelly** of an aircraft which flies at low altitudes (3000-9000m)

AP may be panchromatic or colour and can be used either as map or photo.

AP are suitable for detailed surveying and mapping projects

Difference between SR and AP data collection ?

Aerial Photograph showing a part of UK



Both ***Satellite*** and ***Aerial Photographs*** are subjected to the process of ***Georeferencing*** before being utilized for application purpose.

Toposheets

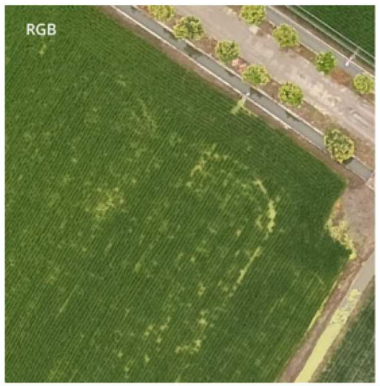
GCPs collected by GPS / DGPS

Drones

RGB-camera

Function: Captures visible light, providing high-resolution color imagery;

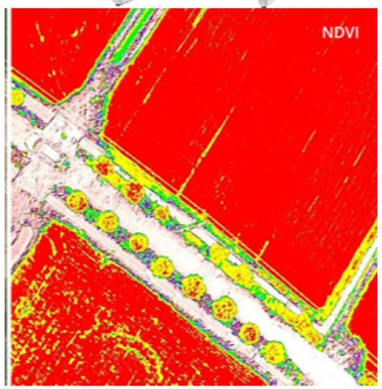
Limitations: Limited to day light conditions



Multi- spectral camera

Function: Captures data across multiple spectral bands, aiding vegetation identification and classfication;

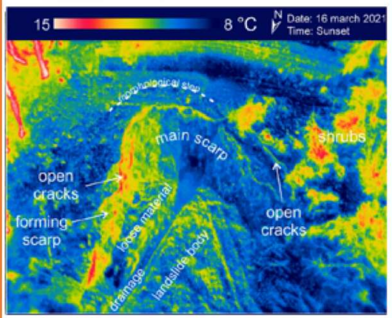
Limitations: Might limited spatial resolution



IR-camera

Function: Detects infrared radiation for identifying temperature;

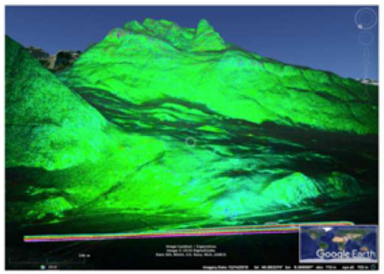
Limitations: Cannot detect geological feature directly; Might limited spatial resolution



SAR

Function: Penetrates through clouds and darkness, providing all weather and day/night monitoring.

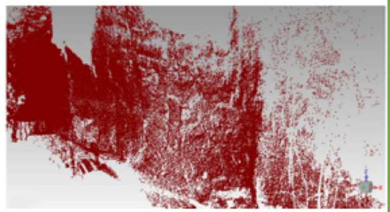
Limitations: Lower spatial resolution, complex data processing

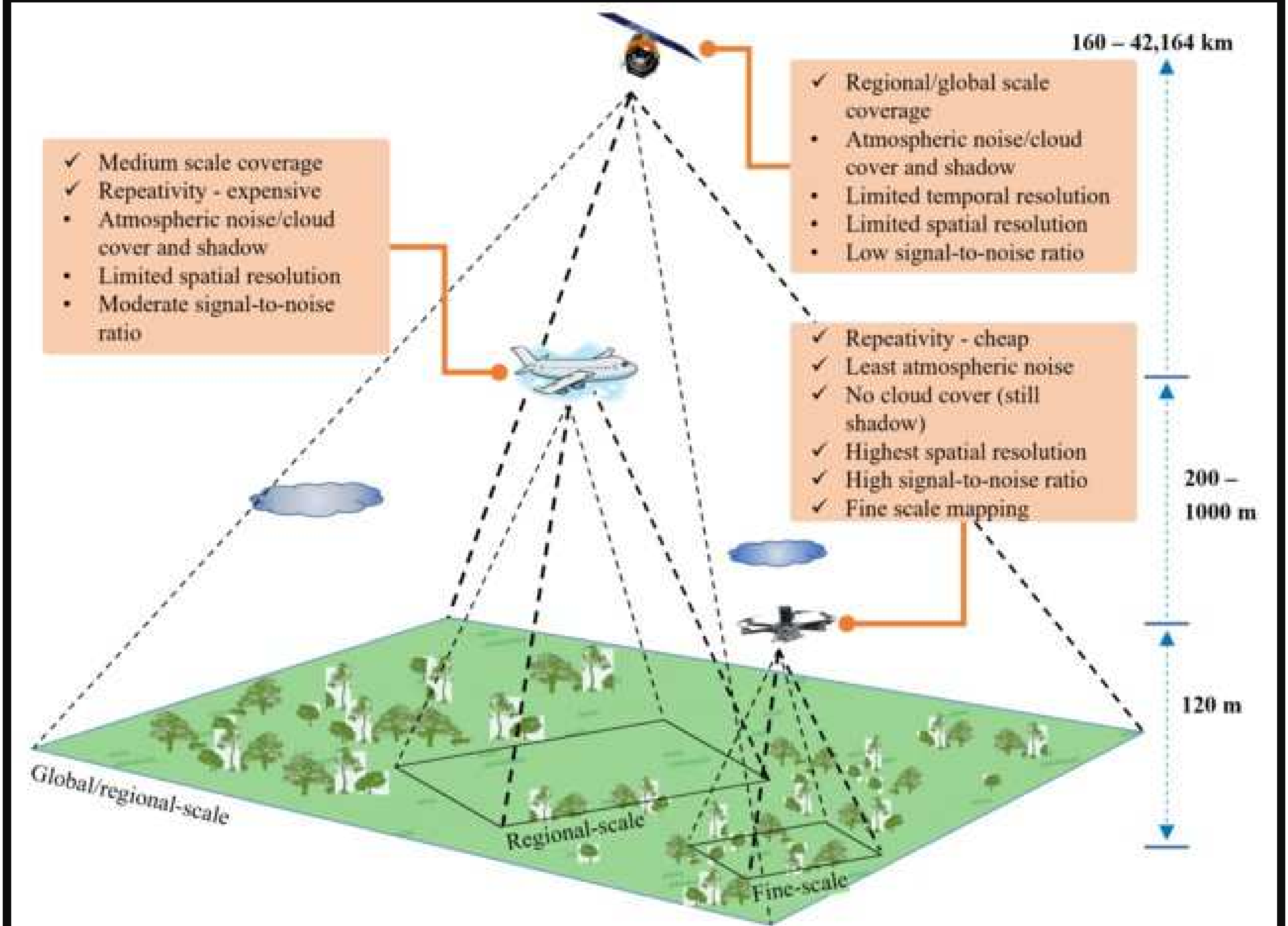


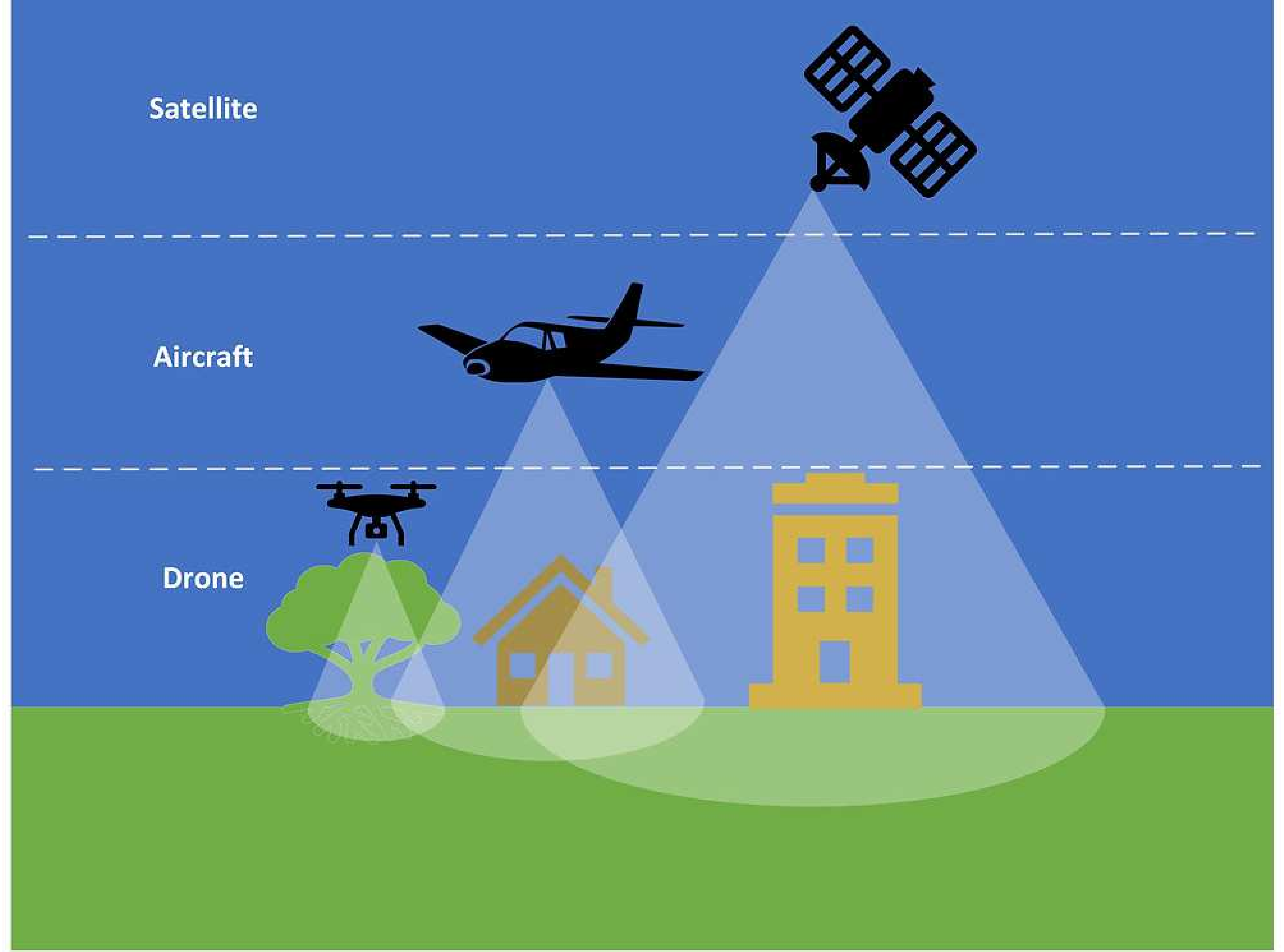
LiDAR

Function: Measures distance using laserpulses creating 3D terrain model directly.

Limitations: Can be affected by atmospheric conditions, cost




160 – 42,164 km
✓ Regional/global scale coverage
• Atmospheric noise/cloud cover and shadow
• Limited temporal resolution
• Limited spatial resolution
• Low signal-to-noise ratio
✓ Medium scale coverage
✓ Repeativity - expensive
• Atmospheric noise/cloud cover and shadow
• Limited spatial resolution
• Moderate signal-to-noise ratio
✓ Repeativity - cheap
✓ Least atmospheric noise
✓ No cloud cover (still shadow)
✓ Highest spatial resolution
✓ High signal-to-noise ratio
✓ Fine scale mapping
200 – 1000 m
120 m
Global/regional-scale
Regional-scale
Fine-scale


Satellite
Aircraft
Drone